Does Mechanism Help? A Prospective Evaluation of a Compartmental SIRS Model Against Statistical Baselines in the CDC FluSight 2026–27 Challenge

Working draft and project log

Bernard Frimpong
Independent researcher (M.S. Applied Mathematics, University of Akron)
bernardfrimpong040@gmail.com

Draft updated: 20 October 2026 (Stage 1 complete)

Project status at a glance

Goal. Compete in CDC FluSight 2026–27 every week (7 Oct 2026 – 26 May 2027) with models of increasing structure, then publish a season-long comparison.
Current stage. Stage 1 complete: data pipeline, flat baseline model, submission validator and a backtest on the 2025–26 season.
Next. First live submission (reference date 24 October 2026, due Wednesday 21 October, 11 PM Eastern); then Stage 2, a statistical growth model.
Repository. https://github.com/bernardfrimpong040-pixel/flu-forecasting-sirs

Abstract

Influenza causes tens of thousands of hospital admissions in the United States each season, and short-term forecasts of those admissions help hospitals and public health agencies plan. The CDC's FluSight challenge collects weekly probabilistic forecasts of laboratory-confirmed influenza hospital admissions for every state from dozens of modelling teams. We ask whether a mechanistic susceptible–infected–recovered–susceptible (SIRS) model, which represents how an epidemic grows and burns out, forecasts these admissions better than statistical models that only extrapolate recent data, and in which phase of the epidemic any advantage appears. All forecasts were submitted to the FluSight hub in real time during the 2026–27 season, before the outcomes were known. [Results: relative weighted interval score of the SIRS model versus the baseline, overall and by epidemic phase; prediction-interval coverage; conclusion.]

1 Introduction

Seasonal influenza places a heavy, variable load on hospitals. In the United States, weekly laboratory-confirmed influenza admissions peaked at 55,544 in the 2024–25 season and 42,500 in 2025–26 (Section 2). Because season timing and size change from year to year, short-term forecasts one to four weeks ahead are a practical planning tool.

Since the 2013–14 season the CDC has run FluSight, a collaborative forecasting challenge in which teams submit forecasts in a common format and are evaluated against the same data (Reich

et al., 2019; Mathis et al., 2024). Models range from statistical time-series methods to mechanistic compartmental models descended from the work of Kermack and McKendrick (1927), some combined with data assimilation (Shaman and Karspeck, 2012).

Comparisons between model families are often made retrospectively, where choices can be tuned, even unintentionally, to outcomes that are already known. This study is *prospective*: every forecast was timestamped in the public FluSight repository before the data it predicts were released.

Research question. Across a full season of real-time forecasts, does a mechanistic SIRS model forecast state-level influenza hospital admissions more accurately than statistical models, and does any advantage depend on the phase of the epidemic (growth, peak, decline)?

Contributions.

1. A transparent, open-source forecasting pipeline submitted weekly to FluSight 2026–27.
2. A sequence of models of increasing structure, from a flat baseline to an SIRS model with time-varying transmission, evaluated on identical weeks.
3. An evaluation stratified by epidemic phase and jurisdiction size.

2 Data

Target. The forecast target is the weekly number of laboratory-confirmed influenza hospital admissions reported to the National Healthcare Safety Network (NHSN) (Centers for Disease Control and Prevention, 2026b), as published in the FluSight hub's `target-data` folder (Centers for Disease Control and Prevention, 2026a). Weeks are CDC epidemiological weeks (Sunday–Saturday) and are labelled by their final Saturday. The data cover 53 jurisdictions (the 50 states, the District of Columbia, Puerto Rico and the national total) from the week ending 5 February 2022. Three states have gaps in 2024 (Massachusetts and Minnesota, May–October; West Virginia, September–October); those weeks are excluded.

Reporting revisions. Recent weeks can be revised as late reports arrive. To evaluate forecasts against the information actually available at the time, a dated snapshot of the target data is saved every time it is downloaded.

Season 2026–27 so far. National weekly admissions rose from 1,247 in the week ending 5 September 2026 to 3,835 in the week ending 10 October 2026, an earlier start than recent seasons (Figure 1). In California and Florida, admissions in week 10 of the season were roughly three to five times their level at the same point of 2025–26.

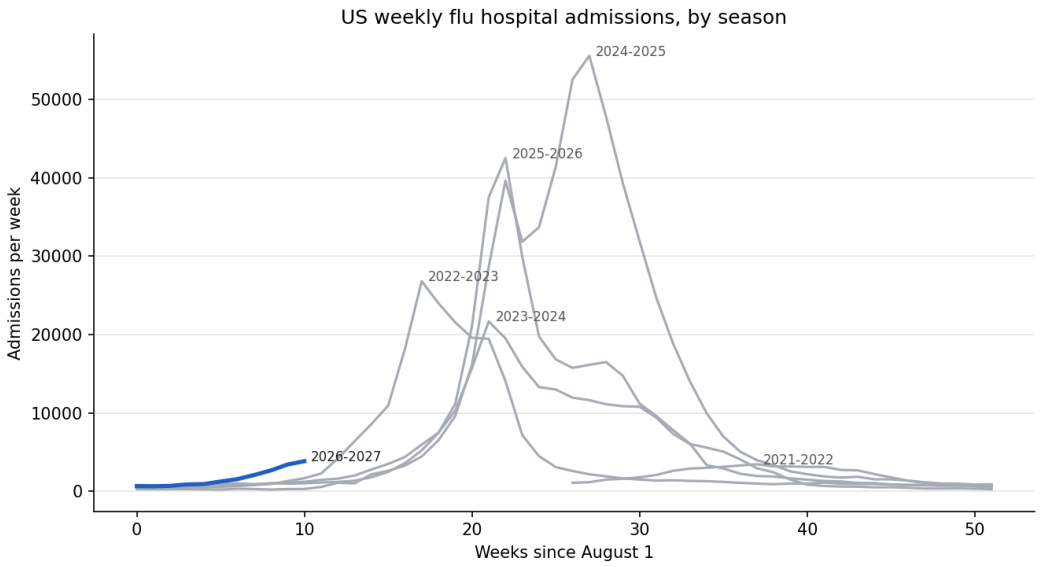


Figure 1: US weekly laboratory-confirmed influenza hospital admissions by season, aligned by weeks since 1 August. The current season (2026–27) is shown in blue.

Table 1: National peak weekly admissions by season.

Season	Peak week ending	Peak admissions
2022–23	3 Dec 2022	26,795
2023–24	30 Dec 2023	21,662
2024–25	8 Feb 2025	55,544
2025–26	3 Jan 2026	42,500
2026–27	*in progress*	

3 Forecasting task

Each Wednesday, teams submit forecasts for the epidemiological week ending on the following Saturday (the *reference date*, horizon $h = 0$) and the three weeks after it ($h = 1, 2, 3$). The newest data available on Wednesday end one week before the reference date, so a forecast at horizon h is made $k = h + 1$ weeks after the last observation. Forecasts are 23 quantiles of the predictive distribution at levels

$$\tau \in \{0.01,\ 0.025,\ 0.05,\ 0.10,\ 0.15,\ \ldots,\ 0.90,\ 0.95,\ 0.975,\ 0.99\},$$

and admission counts must be whole numbers.

4 Methods

4.1 Stage 1: flat log-scale baseline

Let y_t be weekly admissions in one jurisdiction and $z_t = \log(y_t+1)$. The logarithm puts proportional changes on a common scale (a rise from 100 to 120 and from 1,000 to 1,200 are equal), and the $+1$

handles weeks with zero admissions.

The one-week changes $d_t = z_t - z_{t-1}$ are collected for all pairs of consecutive weeks (exactly seven days apart) in the previous 156 weeks, restricted to in-season months (October–May). Each change is also included with its sign reversed, giving a symmetric set $D = \{d_t\} \cup \{-d_t\}$, so the forecast median equals the last observation. Let $\hat{q}_\tau$ be the empirical τ-quantile of D.

For a target k weeks after the last observation y_T, the forecast quantile is

$$\hat{y}_{T+k}(\tau) \;=\; \max\Big(0,\; \mathrm{round}\big(\exp(z_T + \sqrt{k}\,\hat{q}_\tau) - 1\big)\Big). \tag{1}$$

The factor $\sqrt{k}$ comes from treating z_t as a random walk: if weekly changes are independent with variance σ^2, the change over k weeks has variance $k\sigma^2$ and standard deviation $\sqrt{k}\,\sigma$. Jurisdictions are modelled independently.

Planned methods (to be written as each stage is built)

- **Stage 2** – statistical growth model: regression on recent z_t to estimate the current growth rate.
- **Stage 3** – SIRS model, fitted per jurisdiction:

$$\frac{dS}{dt} = -\beta(t)\frac{SI}{N} + \omega R, \qquad \frac{dI}{dt} = \beta(t)\frac{SI}{N} - \gamma I, \qquad \frac{dR}{dt} = \gamma I - \omega R,$$

  with hospital admissions modelled as a fitted fraction of new infections.
- **Stage 4** – uncertainty from an ensemble of fitted parameter sets or a particle filter.
- **Stage 5** (stretch) – hybrid: SIRS plus a learned error correction.

5 Evaluation

Weighted interval score. Following Bracher et al. (2021), a forecast given by quantiles x_τ at the level set $\mathcal{T}$ (with $|\mathcal{T}| = 23$ and the median included) is scored against the observed value y by

$$\mathrm{WIS}(x, y) \;=\; \frac{2}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \big(\mathbf{1}\{y < x_\tau\} - \tau\big)\,(x_\tau - y), \tag{2}$$

the average quantile (pinball) loss, which is a proper scoring rule (Gneiting and Raftery, 2007). Lower is better. WIS rewards forecasts that are both close to the truth and no wider than necessary.

Coverage. The empirical coverage of the central 50% and 95% prediction intervals (between the 0.25 and 0.75, and the 0.025 and 0.975 quantiles). A well-calibrated forecaster achieves about 0.50 and 0.95.

Relative skill. [Season results will be reported as relative WIS versus the FluSight baseline and ensemble, by horizon, by epidemic phase (growth, peak, decline) and by jurisdiction size.]

6 Results

6.1 Retrospective check of the baseline on the 2025–26 season

Before the live season, the baseline (Section 4.1) was run for each of the 34 Saturdays from 11 October 2025 to 30 May 2026 using only data dated before each reference week, and scored against the later observations (Table 2).

Table 2: Baseline backtest on the 2025–26 season. WIS is in admissions per week; "States" averages over 52 jurisdictions. Ideal coverage is 0.50 and 0.95.

Level	Horizon (weeks after last data)	Mean WIS	50% coverage	95% coverage
States	0 (1)	36.3	0.53	0.96
States	1 (2)	66.3	0.46	0.94
States	2 (3)	92.3	0.43	0.91
States	3 (4)	114.0	0.40	0.89
US	0 (1)	1,601.8	0.56	0.91
US	1 (2)	3,170.7	0.44	0.91
US	2 (3)	4,453.1	0.32	0.74
US	3 (4)	5,450.1	0.29	0.74

Two patterns stand out. First, one week ahead the intervals are close to nominal coverage, but coverage falls with horizon, most sharply for the national series (95% coverage of 0.74 at four weeks). The $\sqrt{k}$ scaling in (1) assumes independent weekly changes; during an epidemic, a rise this week tends to be followed by a rise next week, so uncertainty grows faster than $\sqrt{k}$. Second, the flat median lags every turn of the epidemic by one to two weeks (Figure 2), under-predicting the growth phase and over-predicting the decline. A model with epidemic dynamics should reduce exactly this error; testing whether it does is the purpose of Stages 2–3.

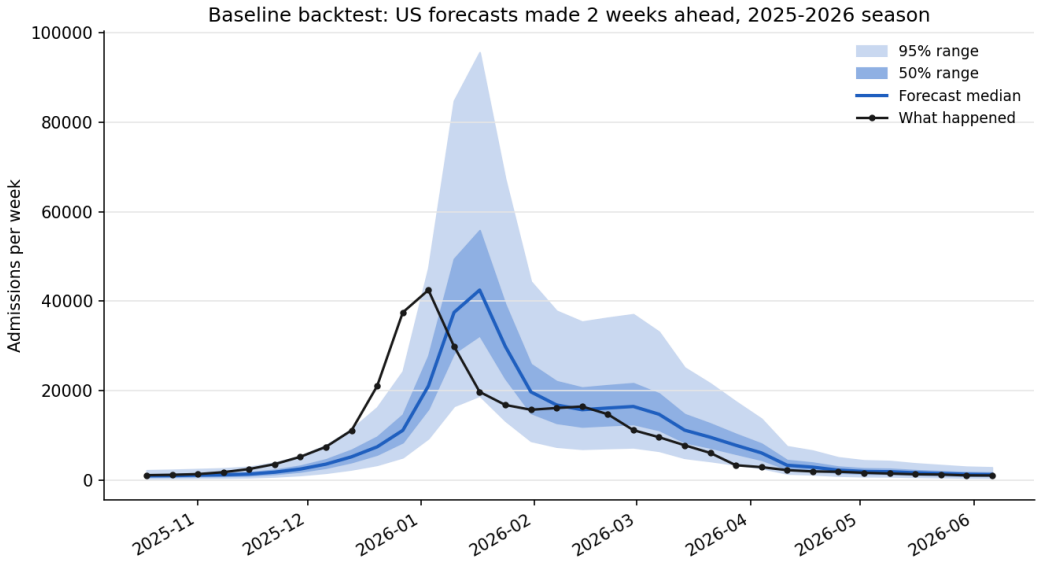


Figure 2: Baseline forecasts for the US made two weeks after the last observation (horizon 1), 2025–26 season: median (line), 50% and 95% intervals (bands), and observed admissions (black).

Limitation of the backtest. It uses the current, revised data rather than the data as first reported, so live performance is expected to be somewhat worse.

6.2 Live season 2026–27

[Weekly results to be added as the season progresses.]

7 Discussion

[To be written after the season: main finding; when mechanism helps or does not; comparison with other FluSight models; limitations (single season, reporting revisions, independent jurisdictions); future work.]

Code and data availability

All code, weekly forecast files and dated data snapshots are available at https://github.com/bernardfrimpong040-pixel/flu-forecasting-sirs. Forecasts were also submitted to the FluSight hub (Centers for Disease Control and Prevention, 2026a). All data are public, aggregated counts; no individual-level data were used.

Use of AI assistance

Code and draft text were prepared with the help of Claude (Anthropic). The author reviewed, ran and verified all code, made the modelling decisions, and takes responsibility for the content.

References

Johannes Bracher, Evan L. Ray, Tilmann Gneiting, and Nicholas G. Reich. Evaluating epidemic forecasts in an interval format. *PLOS Computational Biology*, 17(2):e1008618, 2021.

Centers for Disease Control and Prevention. FluSight 2026–2027 forecast hub. https://github.com/cdcepi/FluSight-forecast-hub, 2026a. Accessed 20 October 2026.

Centers for Disease Control and Prevention. Weekly hospital respiratory data (HRD) metrics by jurisdiction, national healthcare safety network (NHSN). https://data.cdc.gov/Public-Health-Surveillance/Weekly-Hospital-Respiratory-Data-HRD-Metrics-by-Ju/ua7e-t2fy, 2026b. Accessed 20 October 2026.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

William Ogilvy Kermack and Anderson G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A*, 115(772):700–721, 1927.

Sarabeth M. Mathis et al. Evaluation of FluSight influenza forecasting in the 2021–22 and 2022–23 seasons with a new target laboratory-confirmed influenza hospitalizations. *Nature Communications*, 15:6289, 2024.

Nicholas G. Reich, Logan C. Brooks, Spencer J. Fox, et al. A collaborative multiyear, multimodel assessment of seasonal influenza forecasting in the United States. *Proceedings of the National Academy of Sciences*, 116(8):3146–3154, 2019.

Jeffrey Shaman and Alicia Karspeck. Forecasting seasonal outbreaks of influenza. *Proceedings of the National Academy of Sciences*, 109(50):20425–20430, 2012.

A Project log

A.1 Goals

1. Submit a valid FluSight forecast every Wednesday from October 2026 to 26 May 2027.
2. Build models of increasing structure: baseline → statistical growth → SIRS → SIRS with uncertainty → hybrid.
3. Post the season evaluation as a medRxiv preprint and submit to a journal (June–July 2027).

A.2 Stage progress

Stage	Content	Status	Key outputs
1	Data pipeline, flat baseline, validator, backtest	Complete (20 Oct 2026)	Scripts 01–05; Table 2; Figures 1, 2
2	Statistical growth model	Next	–
3	SIRS with time-varying $\beta(t)$	Planned	–
4	Ensemble / particle-filter uncertainty	Planned	–
5	Hybrid SIRS + ML correction	Stretch	–
6	Season evaluation and paper	June 2027	–

A.3 Decisions and reasons

- **Work on the log scale** – epidemic changes are proportional.
- **Use only in-season weeks (Oct–May) for the baseline's changes** – summer weeks with tiny counts would otherwise distort the spread.
- **Keep dated data snapshots** – needed to evaluate against the data as it was on each forecast day.
- **Submit the baseline as a non-designated reference model** – it is a yardstick, not a contender for the CDC ensemble.

A.4 Open questions

- How to define epidemic phases (growth, peak, decline) objectively for the stratified evaluation.
- Whether to forecast the optional targets (rate-trend category, peak week) once the SIRS model is running.
- Before posting the preprint: contact `flusight@cdc.gov` about attribution of hub forecast data.

A.5 Weekly submission log

Reference date	Model(s) submitted	Data through	Notes
24 Oct 2026	BFrimpong-baseline	17 Oct 2026	First submission (planned)